Shayan Hashemi

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Education

2020–Present **Ph.D. in Information Processing Science,** Department of Information Technology and Electrical Engineering, University of Oulu, Finland.

Supervisor: Prof. Mika Mäntylä.

Research Topic: Software Log Anomaly Detection.

2016–2019 **M.Sc. in Computer Engineering, Software,** Department of Electrical and Computer Engineering, Isfahan University of Technology, Iran.

Supervisor: Dr. Mehran Safayani.

Research Topic: Visual Question-Answering.

2010–2016 **B.Sc in Computer Engineering, Software,** Department of Computer, Azad University - Najafabad Branch, Iran.

Project Advisor: Dr. Mehdi Jabalameli.

Publications

S. Hashemi and M. Mäntylä. "OneLog: Towards End-to-End Learning in Software Log Anomaly Detection".

S. Hashemi and M. Mäntylä. "SiaLog: Detecting Anomalies in Software Execution Logs using the Siamese Network" [Link].

S. Hashemi, M. Safayani, and A. Mirzayi. "Multiple answers to a question: a new approach for visual question answering" [Link].

M. Mäntylä, M. Varela, and **S. Hashemi**. "Pinpointing Anomaly Events in Logs from Stability Testing – N-Grams vs. Deep-Learning" [Link].

Articles

Neural Knapsack, *Solving the knapsack problem using neural netwoks*, [Link].

Neural Cryptography, *Neural embedding of a message into an image using deep neural networks*, [Link].

Research Experiences

2020-Present **Doctoral Researcher**, *Software log anomaly detection*, University of Oulu.

2017-2019 **Master's Thesis Researcher**, *Multi-answer visual question answering using deep neural networks*, Isfahan University of Technology.

Teaching Experiences

Spring 2012 Teaching assistant for **Advanced Programming**, *Department of Computer, Azad University - Najafabad Branch, Iran.*

Fall 2012 Teaching assistant for **Principles of Computer and Programming**, *Department of Computer, Azad University - Najafabad Branch, Iran.*

Spring 2018 Teaching assistant for **Operating Systems**, *Department of Electrical and Computer Engineering, Isfahan University of Technology, Iran.*

Fall 2018 Teaching assistant for **Advanced Programming**, *Department of Electrical and Computer Engineering, Isfahan University of Technology, Iran.*

Spring 2022 Teaching assistant for **Next Generation of Software Engineering**, *Department of Information Technology and Electrical Engineering, University of Oulu, Finland.*

Work Experiences

Jan 2020 - Now **AI Researcher**, *University of Oulu*, Software Log Analysis Research.

Dec 2018 - Apr 2019 **Junior AI Researcher**, *Isfahan University of Technology.*

- During two years of research in visual question answering, I introduced a new topic suggesting that a question may have multiple correct answers within a single image. Furthermore, I presented a solution that integrates previous object detection methods with visual question-answering models and utilizes that to produce the answer. My research was published as a paper named "“Multiple answers to a question: a new approach for visual question answering” [Link]."

Mar 2015 - Aug 2016 **Senior Software Engineer**, *Betternet Technologies.*

- I developed the Betternet iOS app from scratch using UIKit and Objective-C ([Link])
- Contributed to the development of Betternet Android app using Java ([Link])
- Member of a team responsible for a new VPN protocol called Hexatech for Linux, iOS, Android using C++, Java, JNI, and Objective-C
- Developed multiple embedded system services for the smart router device called “Betterspot” ([Link]). The router passes the entire traffic of its clients through a VPN tunnel. I used C/C++ for developing the embedded system services, such as switching VPN services and changing the LED color, and Lua for web services.

Jan 2013 - Jan 2014 **Senior Software Engineer**, *Zana Mob Co.*

- I got involved in multiple Android application development projects

May 2010 - Oct 2012 **Junior Software Engineer**, *BASA Co.*

- Responsible for software and service maintenance of the Isfahan province portal
- Developed multiple Liferay-based portlets for the Isfahan province portal
- Contributed to localization of a retail shop management software called “Retail Pro”
- Developed a new website for BASA Co

Certificates

Coursera
- Improving Deep Neural Networks, *Deeplearning.ai*
- Structuring Machine Learning Projects, *Deeplearning.ai*
- Convolutional Neural Networks, *Deeplearning.ai*
- Sequence Models, *Deeplearning.ai*
- Natural Language Processing, *HSE University*
- Sequence, Time Series and Prediction, *Deeplearning.ai*
- Natural Language Processing with Sequence Models, *Deeplearning.ai*
- Natural Language Processing in TensorFlow, *Deeplearning.ai*

Collaborations

- As a freelancer, I undertook a natural language processing software project to detect spam emails in Persian. I was responsible for designing, implementing, and deploying an NLP model to analyze email contents and detect spam.
- As a freelancer, I developed customer and collector applications for iOS and Android for recycling collection companies. I used Java to implement the Android and Use UIKit and Swift for iOS.
- An intelligent software for pinpointing anomalous log lines in large stability test logs. I was responsible for parsing log messages into machine-processable vectors.

References

- References are available upon request.